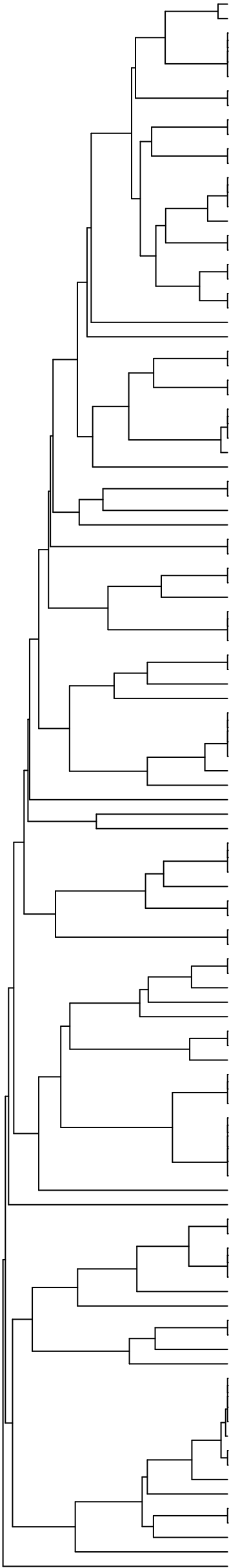


- 14/333 obsolete organic hydroxy compound metabolic process  
4/56 multicellular organismal response to stress  
2/25 detection of stimulus involved in sensory perception of pain  
8/93 detection of stimulus involved in sensory perception  
12/161 sensory perception of mechanical stimulus  
20/361 sensory perception  
34/651 nervous system process  
2/23 detection of mechanical stimulus involved in sensory perception of pain  
4/44 detection of mechanical stimulus involved in sensory perception  
2/19 detection of mechanical stimulus involved in sensory perception of touch  
14/153 response to mechanical stimulus  
5/61 detection of mechanical stimulus  
4/32 sensory perception of touch  
1/30 detection of light stimulus  
10/135 detection of abiotic stimulus  
5/55 detection of visible light  
12/206 detection of stimulus  
5/92 sensory perception of chemical stimulus  
5/84 positive regulation of transporter  
4/181 positive regulation of transporter  
4/37 positive regulation of cation channel  
4/74 obsolete regulation of cation channel  
6/19 positive regulation of monoatomic ion transmembrane transport  
7/174 regulation of monoatomic ion transmembrane transport  
2/20 regulation of membrane repolarization  
2/22 response to magnesium ion  
12/151 negative regulation of cytosolic calcium ion concentration  
12/157 regulation of cytosolic calcium ion concentration  
3/18 positive regulation of voltage-gated potassium channel  
17/229 locomotory behavior  
11/239 regulation of membrane potential  
6/13 action potential  
2/18 cell-cell signaling involved in cardiac conduction  
4/60 obsolete multicellular organismal signaling  
4/67 transmission of nerve impulse  
10/115 mating  
53/782 behavior  
5/64 courting behavior  
2/43 male courting behavior  
2/196 reproductive behavior  
3/23 male mating behavior  
2/18 mating behavior, sex discrimination  
5/26 regulation of vasoconstriction  
7/121 regulation of tube diameter  
7/142 regulation of tube size  
3/35 membrane depolarization  
1/44 regulation of postsynaptic membrane potential  
0/19 gamma-aminobutyric acid signaling pathway  
12/267 G protein-coupled receptor signaling pathway  
10/124 chemosensory behavior  
4/60 obsolete multi-organism behavior  
2/25 ecological process involved in intraspecies interaction between organisms  
2/11 neural crest cell fate commitment  
4/751 central nervous system development  
2/21 neural crest formation  
4/17 neurectodermal development  
40/609 cell-cell signaling  
40/645 embryonic morphogenesis  
14/217 gastrulation  
1/45 neuroectodermal cell migration  
40/717 tube morphogenesis  
32/554 circulatory system development  
2/21 lymph vessel development  
3/24 face development  
11/243 imaginal disc-derived appendage development  
20/407 instar larval or pupal development  
4/81 leg disc development  
45/773 tissue morphogenesis  
19/336 appendage development  
19/405 obsolete post-embryonic animal organ development  
38/701 post-embryonic development  
11/261 wing disc development  
16/397 imaginal disc development  
33/647 sensory organ development  
14/372 sensory organ morphogenesis  
5/67 somatic muscle development  
3/93 cuticle development  
38/798 cell morphogenesis  
1/14 axon target recognition  
7/134 axon development  
16/244 neuron projection guidance  
27/410 taxis  
2/119 neurite projection extension  
35/656 obsolete cell morphogenesis involved in differentiation  
12/25 developmental cell growth  
5/108 neuron recognition  
6/151 cell recognition  
3/17 regulation of protein localization to synapse  
18/462 regulation of anatomical structure size  
14/283 regulation of cellular component size  
30/648 regulation of anatomical structure morphogenesis  
10/245 obsolete regulation of cell morphogenesis involved in differentiation  
13/346 regulation of cell morphogenesis  
6/157 regulation of axonogenesis  
20/443 regulation of developmental growth  
30/658 regulation of growth  
1/9 negative regulation of cardiac muscle hypertrophy  
2/14 long-term synaptic depression  
2/654 regulation of cell projection organization  
0/22 positive regulation of lamellipodium organization  
17/394 positive regulation of cell projection organization  
0/36 regulation of lamellipodium organization  
6/155 regulation of cell projection assembly  
3/67 positive regulation of plasma membrane bounded cell projection assembly  
2/18 Ras protein signal transduction  
28/554 obsolete regulation of cellular component movement  
7/174 negative regulation of locomotion  
37/658 regulation of locomotion  
9/568 actin filament-based process  
2/44 actin filament organization  
8/267 supramolecular fiber organization  
2/3 actin filament-based movement  
3/21 epithelial cell morphogenesis  
14/263 epithelial cell development  
24/520 epithelial cell differentiation  
5/4 membrane invagination  
5/42 phagocytosis, engulfment  
4/53 mushroom body development  
3/44 positive regulation of Rho protein signal transduction  
4/64 regulation of Rho protein signal transduction  
4/183 regulation of small GTPase mediated signal transduction  
3/21 positive regulation of small GTPase mediated signal transduction  
7/24 regulation of body fluid levels  
32/641 microtubule-based process  
10/130 cilium movement  
17/246 microtubule-based movement  
5/102 microtubule bundle formation  
27/755 cytoskeleton organization  
2/18 inner dynein arm assembly  
2/1 axonemal dynein complex assembly  
42/715 localization of cell  
20/398 cell projection assembly  
2/29 intracellular transport involved in cilium assembly  
7/70 microtubule-based transport  
2/44 intracellular transport  
1/44 regulation of protein depolymerization  
1/26 regulation of microtubule depolymerization  
1/35 negative regulation of protein-containing complex disassembly  
1/13 distal tubule development  
1/83 leukocyte migration  
7/141 cellular response to radiation  
19/578 DNA metabolic process  
5/68 skeletal muscle cell differentiation  
4/55 aminoglycan catabolic process  
6/116 aminoglycan metabolic process  
3/28 cytochrome complex assembly  
2/20 respiratory chain complex IV assembly  
3/70 proton transmembrane transport  
19/340 mitochondrion organization  
3/31 inner mitochondrial membrane organization  
9/163 mitochondrial transport  
1/12 obsolete negative regulation of necrotic cell death  
2/88 oxidative phosphorylation  
6/135 cellular respiration  
6/135 energy derivation by oxidation of organic compounds  
6/53 ATP synthesis coupled electron transport  
6/53 electron transport chain  
15/472 nucleotide metabolic process  
0/11 mitochondrial electron transport, ubiquinol to cytochrome c  
6/176 nucleoside triphosphate metabolic process  
6/176 nucleoside diphosphate metabolic process  
4/68 mitochondrial respiratory chain complex assembly  
0/17 NAADH dehydrogenase complex assembly  
16/532 nucleobase-containing small molecule metabolic process  
0/28 cyclic purine nucleotide metabolic process  
18/712 RNA processing  
8/392 mRNA processing  
18/545 mRNA metabolic process  
1/26 termination of RNA polymerase II transcription  
1/52 DNA-templated transcription termination  
3/147 RNA 3'-end processing  
9/218 protein-RNA complex organization  
8/281 regulation of mRNA metabolic process  
13/399 post-transcriptional regulation of gene expression  
0/22 maturation of 5.8S rRNA from transcribed RNA transcript (SSU-rRNA, 5.8S rRNA, LSU-rRNA)  
0/35 maturation of 5.8S rRNA  
1/56 rRNA metabolic process  
9/416 obsolete ncRNA metabolic process  
6/301 obsolete ncRNA processing  
9/416 ncRNA 3'-end processing  
6/234 ribosome biogenesis  
2/66 ribosomal large subunit biogenesis  
3/66 regulatory ncRNA processing  
5/94 mitochondrial gene expression  
15/55 mitochondrial translation  
20/419 amide biosynthetic process  
5/62 translational elongation  
7/106 translational initiation  
18/384 peptide biosynthetic process  
18/384 peptide metabolic process  
26/612 amide metabolic process  
3/68 de novo protein folding  
12/161 protein folding  
2/19 obsolete chaperone factor-dependent protein refolding  
4/53 obsolete chaperone-mediated protein folding  
3/26 regulation of cellular response to heat  
1/10 regulation of protein localization to chromosome, telomeric region  
6/29 protein import  
2/7781 protein transport  
6/234 establishment of protein localization to organelle  
8/234 nuclear transport  
3/109 import into nucleus  
6/135 nuclear export

p < 0.001  
p < 0.01  
p < 0.05

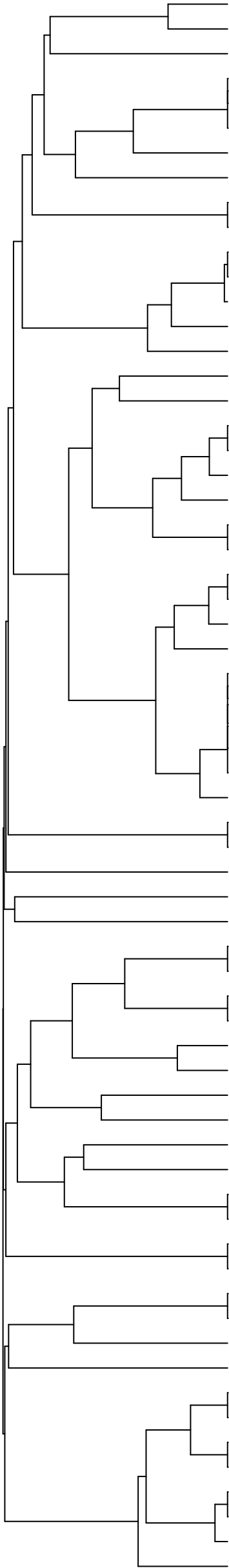
WINTER – Biological Process (BP)



- 14/305 presynapse
  - 7/143 neuron projection terminus
  - 12/294 obsolete axon part
  - 22/477 axon
  - 2/18 paranode region of axon
  - 2/48 main axon
  - 24/568 synapse
    - 4/90 neuromuscular junction
    - 5/139 synaptic membrane
    - 3/70 presynaptic membrane
    - 3/20 obsolete intrinsic component of synaptic membrane
    - 33/606 obsolete intrinsic component of plasma membrane
    - 29/639 somatodendritic compartment
    - 20/432 dendritic tree
    - 7/129 neuron spine
    - 3/57 dendritic shaft
    - 24/531 cell body
    - 3/37 neuronal cell body membrane
    - 10/260 postsynapse
    - 4/98 postsynaptic membrane
      - 1/11 postsynaptic specialization membrane
      - 9/152 neuron to neuron synapse
      - 7/179 membrane microdomain
      - 10/206 plasma membrane protein complex
      - 0/10 lamellipodium membrane
      - 6/112 lamellipodium
      - 9/232 cell leading edge
      - 1/86 ruffle
      - 2/40 neuron projection membrane
      - 5/183 cell projection membrane
      - 24/563 plasma membrane region
      - 3/91 leading edge membrane
      - 1/52 filopodium
      - 16/312 apical part of cell
      - 12/202 apical plasma membrane
      - 1/10 basal cortex
      - 5/128 cluster of actin-based cell projections
  - 25/309 cell surface
    - 5/95 external side of plasma membrane
    - 9/147 adherens junction
    - 15/287 anchoring junction
      - 4/53 cell-cell contact zone
      - 5/70 apical junction complex
      - 2/16 septate junction
      - 3/49 tight junction
    - 4/213 actin cytoskeleton
      - 0/48 actin filament
      - 7/202 polymeric cytoskeletal fiber
      - 0/53 cortical cytoskeleton
      - 6/75 i band
      - 8/111 sarcomere
      - 9/153 contractile muscle fiber
      - 15/352 supramolecular fiber
      - 0/10 myofilament
      - 1/13 costamere
      - 3/100 extrinsic component of plasma membrane
    - 1/19 protein folding chaperone complex
    - 6/129 myelin sheath
    - 1/42 vesicle coat
    - 2/73 membrane coat
      - 23/563 membrane protein complex
      - 6/107 coated vesicle membrane
      - 2/65 Golgi-associated vesicle membrane
      - 4/126 Golgi-associated vesicle
      - 2/94 specific granule
      - 26/584 secretory vesicle
      - 2/22 intracellular transport particle
    - 24/459 cilium
      - 7/115 ciliary basal body
      - 1/58 ciliary transition zone
      - 6/117 non-motile cilium
      - 5/95 9+2 motile cilium
      - 9/136 motile cilium
      - 9/130 obsolete sperm part
      - 6/49 obsolete axoneme part
      - 8/139 plasma membrane bounded cell projection cytoplasm
      - 15/381 cytoplasmic region
      - 9/148 microtubule associated complex
      - 4/31 axonemal dynein complex
      - 34/763 microtubule cytoskeleton
      - 6/57 dynein complex
      - 4/20 inner dynein arm
      - 0/21 centriolar satellite
      - 0/14 guanylate cyclase complex, soluble
      - 1/14 cytoplasmic side of rough endoplasmic reticulum membrane
      - 5/93 cytosolic ribosome
      - 5/55 small ribosomal subunit
      - 7/160 ribosome
      - 22/674 ribonucleoprotein complex
        - 1/24 rough endoplasmic reticulum membrane
        - 16/667 nucleolus
        - 0/82 Sm-like protein family complex
        - 0/17 U1 snRNP
        - 0/21 prespliceosome
        - 0/10 methylosome
        - 1/59 respiratory chain complex
        - 25/470 mitochondrial envelope
          - 0/26 NADH dehydrogenase complex
          - 30/757 organelle envelope
          - 5/143 obsolete mitochondrial membrane part
          - 3/79 inner mitochondrial membrane protein complex
          - 12/326 organelle inner membrane
          - 6/176 mitochondrial protein-containing complex
          - 0/18 cytochrome complex
          - 33/690 obsolete mitochondrial part
          - 11/343 mitochondrial matrix
          - 0/78 oxidoreductase complex
          - 7/80 organelle envelope lumen
          - 3/40 histone deacetylase complex

p < 0.001  
p < 0.01  
p < 0.05

## WINTER – Cellular Component (CC)



8/118 cell adhesion molecule binding

43/672 signaling receptor binding

10/372 GTPase binding

16/468 cytoskeletal protein binding

3/160 actin binding

1/74 actin filament binding

1/12 spectrin binding

4/42 kinesin binding

9/336 lipid binding

7/239 phospholipid binding

8/72 microtubule motor

9/108 polypeptide conformation or assembly isomerase

11/224 macromolecular conformation isomerase

7/43 dynein light intermediate chain binding

6/41 dynein intermediate chain binding

5/45 scaffold protein binding

3/23 actinin binding

6/27 sodium channel

12/155 monoatomic cation channel

14/219 metal ion transmembrane transporter

8/66 ligand-gated monoatomic cation channel

3/18 outward rectifier potassium channel

6/69 voltage-gated monoatomic ion channel

2/33 ligand-gated monoatomic anion channel

6/80 obsolete inorganic anion transmembrane transporter

0/12 GABA-A receptor

0/16 GABA receptor

1/29 extracellular ligand-gated monoatomic ion channel

16/222 passive transmembrane transporter

1/18 excitatory extracellular ligand-gated monoatomic ion channel

9/94 ligand-gated monoatomic ion channel

12/140 gated channel

1/18 transmitter-gated monoatomic ion channel

1/12 neurotransmitter:sodium symporter

1/23 neurotransmitter transmembrane transporter

9/221 protein serine/threonine kinase

1/53 acetyltransferase

0/17 antigen binding

5/87 translation factor, RNA binding

2/47 translation initiation factor

27/682 RNA binding

0/11 mRNA regulatory element binding translation repressor

9/158 GTPase

3/27 translation elongation factor

2/43 rRNA binding

5/125 structural constituent of ribosome

2/93 single-stranded DNA binding

6/33 DNA secondary structure binding

13/483 catalytic, acting on a nucleic acid

8/320 catalytic, acting on RNA

1/27 phosphorus-oxygen lyase

0/14 guanylate cyclase

4/34 ATPase regulator

3/24 ATPase activator

5/79 unfolded protein binding

2/84 obsolete protein transporter

3/75 active monoatomic ion transmembrane transporter

6/123 primary active transmembrane transporter

1/38 oxidoreduction-driven active transmembrane transporter

2/83 proton transmembrane transporter

4/78 oxidoreductase, acting on NAD(P)H

0/27 NADH dehydrogenase

0/34 oxidoreductase, acting on NAD(P)H, quinone or similar compound as acceptor

4/88 electron transfer

p < 0.001

p < 0.01

p < 0.05

WINTER – Molecular Function (MF)